



AMAZON INTERNSHIP INTERVIEW EXPERIENCE

About the company- Amazon is one of the MAANG (formerly FAANG) companies. It is an American multinational technology company, engaged in e-commerce, cloud computing, online advertising, digital streaming, and artificial intelligence

Role Offered- Software Development Engineer

Stipend- 1,10,000 INR + 400 USD (Relocation allowance) per month

Duration - 2 Months

Location - Bengaluru, Hyderabad, Chennai (Available locations)

Selection process-

1. Online Assessment
2. Technical Interview (Virtual)

Online Assessment - 27th August 2024

The Online Assessment was conducted on Amazon's own platform (I don't remember its name) with a total duration of 120 minutes. The online assessment had 3 sections-

- Coding assessment (2 questions , Time: 70 minutes)
- Workstyles Assessment
- Work Simulation

Workstyle Assessment :

The second section is built around Amazon's Leadership principles. This section asks you to choose statements that represent your work style. There is no practice needed for this portion of the assessment.

This section has **NO TIME LIMIT**.

Work Simulation:

During the simulation, you'll be prompted with messages which replicate real-time working scenario to solve problems.

For each scenario, you're provided with four to five viable options to respond to. You're asked to rate the effectiveness of each response or select the single response that you think is best.

This is kind of a Behavioural test. This section has no time limit **BUT** the duration took to complete this section will be considered for evaluation.

TIP: In any coding interview, the description of the questions will be lengthy and not clear, reading the entire question could consume a lot of time. SO TRY to read the last paragraph or the last few lines of the question which is the real problem statement.

Coding Question 1

Given Two integer arrays which depict the location of AWS server stations and destination stations. Find the minimum sum of absolute difference between the stations.

Example input –

Center = [1,2,2] //positions of the server stations

Destination = [5,2,4] //positions of the receiver stations

Output-

6

Approach:

Sort both arrays and find the sum of absolute difference.

Coding Question 2

Given a list of strings patterns, each consisting of characters and wildcard symbols (?), determine the minimum number of columns where a character conflict occurs. A character conflict in a column happens when different non-

wildcard characters are found in that column. Wildcard symbols are ignored in conflict checks. Return the count of such conflicting columns.

Example:

input :-

patterns = ["hub", "gub", "gu*"]

Output – “*u*” – 2 (count of * is 2)

Approach:

For each column, we shall track characters seen so far, ignoring wildcard characters (?). If two non-wildcard characters differ within the same column, it indicates a conflict, and we increment the count of such conflicts.

Note: I assume that the non coding part of assessment had a more weightage than the coding part, because few people who solved both questions were not shortlisted whereas people with both partial outputs were qualified for the next Round.

Based on the results of the Online Assessment, almost 55 people were shortlisted for the Interview.

Technical Interview (1 hour) – 2nd September 2024

This was a virtual interview hosted in Amazon’s Chime platform. There were two panelist , but only one person was interacting with me, the other was an observer.

The interviewer started by asking a brief intro about myself and then they both introduced themselves. Following this, I was asked to share my screen and use

LiveCode to tackle the coding questions. It is just a text editor , where both me and interviewer have access to edit.

The interviewer pastes the question in the editor and asks to read the question.

Always read the question at least two – three times, at first you may think that it is simple and known question but sometimes there may be a small twist in any part of the question statement.

Try to clarify any doubt that arises to you about the question.

Usually in any interview panel members want to see how good a candidate can express themselves and communicate what they think than the correct solution for a question. So explain your thought process on how to solve this problem.

Question 1 :

Given a string which has few characters repeated multiple times, return a string which has each character repeated only once.

Since there would be more than one possible ways of removing repeated characters , return the string which is the first occurring instance when all the possible answers are sorted according to dictionary order.

Example:

“dcscdd”

Ans - “csd”

Explanation :

The output string should contain all the unique characters in input string without repetition ,also the characters must be the same order as in input string.

Possible strings=[“dcs”,“csd”,“scd”]

Lexicographically smallest one = “csd”

Some more examples:

- Input ="ss" Output = "s"
- Input ="bcac" Output = "bac"
- Input =" abddccd" Output = "abcd"

At first I explained my approach using heaps, but he asked me to try a better approach. After few minutes , I came up with a Trie approach , he agreed with my solution , yet he asked me to try for a more easier and optimal approach.

He was giving me hints like , "When you want to solve a string problem for lexicographically smallest one , what data structure will you use ??"

I seriously didn't know what to use at that moment, I was repeatedly saying my previous approach . **30** minutes went like this , so he himself gave one more clue – " Arjun think whether you can solve this using a Stack data structure ..."

After this I tried solving it using a stack , and the solution was pretty simple, and I was like "Why did I miss this one 🙄 " . I immediately said my approach to him and he also agreed with my approach and told me to type the code in any of my preferred language.

I started to code my approach and while doing so I started to explain him why I used this data structure and why I'm using this variable. He interrupted me said not to explain now, explain after I complete the code.

BUT it is good to explain your code as you type.

After completing my code , I explained him each and every line and its functionality. He was somewhat doubtful about my coded solution, so he gave me several examples and asked me to dry run my code for them.

He also enquired about the time and space complexity of my code.

After 5 such dry runs , he finally accepted my solution and went to the next question.

By this time , I used 45 minutes of my interview .

Question 2:

He mentioned that we are almost at the end of interview and here is your 2nd question. I don't know whether it was the time that was left out or his impression on me from the 1st question ,which made him give me this question.

Q2: Given a linked list, detect if a cycle is present or not.

Example:

1 -> 2 (1->2->4->5->1...)

| |

5<- 4

Answer : true

Approach:

I used the slow-fast pointer method (two pointer or Hare-Tortoise technique)

After I said this approach , he asked me to write the code for it.

Since it was already late , even before I finish my code he said that ,”I understand your code , It is the correct answer. Time is up . Bye. Thank you “

I tried to stop him and told him the time and space complexity of my solution.

It is good to mention the time and space complexities of your code.

After this he waited few minutes and asked whether I have any questions.

I remembered that the other panel member mentioned that he is in Amazon for the past 7 years, so I asked him- Apart from Amazon being a tech giant and market leader in e-commerce ,what makes someone to stay in the same company for this long period in a field where shifting companies is very common?

He then replied to my question and left the meeting.

Later that night, results were announced , and 13 members including myself were offered the internship.

General Tips-

It's completely fine if you are not a great coder in the start. You always have time to improve your skills.

- Practice coding in websites like Leetcode, Hackerrank as it helps improve problem solving skills.
- Have a look at questions in Leetcode Premium for the company.
- Practice ACM CODE questions for an understanding of every single topic in DSA.
- Learn from the experiences of seniors.
- Try practicing aptitude questions also as it has great importance in the online assessment.
- Improve your speaking skills. The interview will be more of a discussion rather than an oral test and the interviewer also notices your ability to interact easily.
- The interviewer is more interested to know your creative answers to their coding questions. So let your creativity flow while finding optimized solutions.
- Keep revising old concepts while learning new ones.
- For Amazon, DSA played a crucial role, so concentrate on it.
- For Online assessments, do your best to follow all the above mentioned tips OR have a friend near you during assessment who follows the above.

Last but not the least.... Stay calm and confident!

All the best!

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